

```
-- =====  
  
-- MySQL VIEW DEMO SCRIPT  
  
-- Features:  
  
-- Simple View  
  
-- Complex View  
  
-- Join View  
  
-- Aggregation View  
  
-- View with WHERE  
  
-- View with WITH CHECK OPTION  
  
-- Updatable View  
  
-- Read-only View  
  
-- OR REPLACE View  
  
-- ALGORITHM option  
  
-- SQL SECURITY option  
  
-- DROP View  
  
-- =====
```

```
CREATE DATABASE view_demo;
```

```
USE view_demo;
```

```
-- =====  
  
-- 1. CREATE TABLES  
  
-- =====
```

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
dept_name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY AUTO_INCREMENT,  
    emp_name VARCHAR(100) NOT NULL,  
    gender VARCHAR(10),  
    salary DECIMAL(10,2),  
    dept_id INT,  
    status VARCHAR(20),  
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
);
```

```
-- =====  
-- 2. INSERT SAMPLE DATA  
-- =====
```

```
INSERT INTO departments (dept_name) VALUES  
( 'HR' ),  
( 'IT' ),  
( 'Finance' ),  
( 'Sales' );
```

```
INSERT INTO employees (emp_name, gender, salary, dept_id, status) VALUES  
( 'Aung Aung', 'Male', 800000, 1, 'Active' ),  
( 'Su Su', 'Female', 950000, 2, 'Active' ),
```

```
('Kyaw Kyaw', 'Male', 700000, 2, 'Inactive'),  
('Hla Hla', 'Female', 1200000, 3, 'Active'),  
('Mg Mg', 'Male', 600000, 4, 'Active');
```

```
-- =====
```

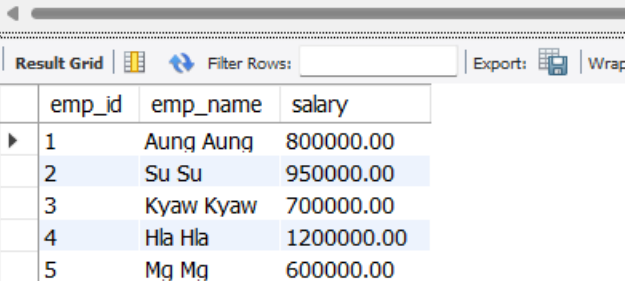
```
-- 3. SIMPLE VIEW
```

```
-- Shows selected columns only
```

```
-- =====
```

```
CREATE VIEW vw_employee_basic AS  
SELECT emp_id, emp_name, salary  
FROM employees;
```

```
66 • SELECT * FROM vw_employee_basic;
```



	emp_id	emp_name	salary
▶	1	Aung Aung	800000.00
	2	Su Su	950000.00
	3	Kyaw Kyaw	700000.00
	4	Hla Hla	1200000.00
	5	Mg Mg	600000.00

```
-- =====
```

```
-- 4. VIEW WITH WHERE CONDITION
```

```
-- Shows only active employees
```

```
-- =====
```

```
CREATE VIEW vw_active_employees AS  
SELECT emp_id, emp_name, salary, status  
FROM employees
```

WHERE status = 'Active';

78 • **SELECT** * **FROM** vw_active_employees;

Result Grid | Filter Rows: | Export: | Wrap Cell C

	emp_id	emp_name	salary	status
▶	1	Aung Aung	800000.00	Active
	2	Su Su	950000.00	Active
	4	Hla Hla	1200000.00	Active
	5	Mq Mq	600000.00	Active

-- =====

-- 5. JOIN VIEW

-- Combines employees and departments

-- =====

CREATE VIEW vw_employee_department AS

SELECT

e.emp_id,

e.emp_name,

e.salary,

d.dept_name

FROM employees e

INNER JOIN departments d

ON e.dept_id = d.dept_id;

95 • **SELECT** * **FROM** vw_employee_department;

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

	emp_id	emp_name	salary	dept_name
▶	1	Aung Aung	800000.00	HR
	2	Su Su	950000.00	IT
	3	Kyaw Kyaw	700000.00	IT
	4	Hla Hla	1200000.00	Finance
	5	Mq Mq	600000.00	Sales

```
-- =====
-- 6. AGGREGATE VIEW
-- Shows department salary summary
-- This view is read-only
-- =====
```

```
CREATE VIEW vw_department_salary_summary AS
```

```
SELECT
```

```
    d.dept_name,
```

```
    COUNT(e.emp_id) AS total_employees,
```

```
    SUM(e.salary) AS total_salary,
```

```
    AVG(e.salary) AS average_salary,
```

```
    MAX(e.salary) AS highest_salary,
```

```
    MIN(e.salary) AS lowest_salary
```

```
FROM departments d
```

```
LEFT JOIN employees e
```

```
ON d.dept_id = e.dept_id
```

```
GROUP BY d.dept_name;
```

```
116 • SELECT * FROM vw_department_salary_summary;
```

```
117
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 						
	dept_name	total_employees	total_salary	average_salary	highest_salary	lowest_salary
▶	HR	1	800000.00	800000.000000	800000.00	800000.00
	IT	2	1650000.00	825000.000000	950000.00	700000.00
	Finance	1	1200000.00	1200000.000000	1200000.00	1200000.00
	Sales	1	600000.00	600000.000000	600000.00	600000.00

```
-- =====  
  
-- 7. VIEW WITH ORDER BY  
  
-- Note: ORDER BY in view is not always guaranteed  
  
-- unless used with LIMIT or ordered again in SELECT  
  
-- =====
```

```
CREATE VIEW vw_high_salary_employees AS  
  
SELECT emp_id, emp_name, salary  
  
FROM employees  
  
WHERE salary >= 800000  
  
ORDER BY salary DESC;
```

130 • **SELECT * FROM** vw_high_salary_employees;
131

Result Grid | Filter Rows: | Export: | Wrap Cell Content

	emp_id	emp_name	salary
▶	4	Hla Hla	1200000.00
	2	Su Su	950000.00
	1	Aung Aung	800000.00

```
-- =====  
  
-- 8. UPDATABLE VIEW  
  
-- Simple view from one table can be updated  
  
-- =====
```

```
CREATE VIEW vw_employee_salary_update AS  
  
SELECT emp_id, emp_name, salary  
  
FROM employees;
```

-- Update data through view

UPDATE vw_employee_salary_update

SET salary = 850000

WHERE emp_id = 1;

146 • **SELECT * FROM employees;**

147

Result Grid | Filter Rows: | Edit: | Export/Imp

	emp_id	emp_name	gender	salary	dept_id	status
▶	1	Aung Aung	Male	850000.00	1	Active
	2	Su Su	Female	950000.00	2	Active
	3	Kyaw Kyaw	Male	700000.00	2	Inactive
	4	Hla Hla	Female	1200000.00	3	Active
	5	Mq Mq	Male	600000.00	4	Active
*	NULL	NULL	NULL	NULL	NULL	NULL

-- =====

-- 9. VIEW WITH CHECK OPTION

-- Prevents inserting/updating rows that violate view condition

-- =====

CREATE VIEW vw_active_employee_check AS

SELECT emp_id, emp_name, salary, status

FROM employees

WHERE status = 'Active'

WITH CHECK OPTION;

-- Valid update

UPDATE vw_active_employee_check

SET salary = 1000000

WHERE emp_id = 2;

164 • **SELECT * FROM** vw_active_employee_check;

165

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	emp_id	emp_name	salary	status
▶	1	Aung Aung	850000.00	Active
	2	Su Su	1000000.00	Active
	4	Hla Hla	1200000.00	Active
	5	Mq Mq	600000.00	Active

-- Invalid update example

-- This will fail because status becomes Inactive

UPDATE vw_active_employee_check

SET status = 'Inactive'

WHERE emp_id = 2;

-- error message

-- Error Code: 1369. CHECK OPTION failed 'view_demo.vw_active_employee_check'

-- =====

-- 10. CASCADED CHECK OPTION

-- Default behavior

-- Checks conditions of this view and underlying views

-- =====

CREATE OR REPLACE VIEW vw_high_paid_active AS

SELECT emp_id, emp_name, salary, status

FROM vw_active_employee_check

WHERE salary >= 800000

WITH CASCADED CHECK OPTION;

188 • **SELECT * FROM** vw_high_paid_active;

189

Result Grid | Filter Rows: | Export: | Wrap Cell

	emp_id	emp_name	salary	status
▶	1	Aung Aung	850000.00	Active
	2	Su Su	1000000.00	Active
	4	Hla Hla	1200000.00	Active

-- don't update, follow underlying Check Option (stauts=active)

UPDATE vw_high_paid_active

SET salary = 999

WHERE emp_id = 3;

UPDATE vw_high_paid_active

SET salary = 999

WHERE emp_id = 1;

-- Error Code: 1369. CHECK OPTION failed 'view_demo.vw_high_paid_active'

-- check current Check Option and underlying check option

-- =====

-- 11. LOCAL CHECK OPTION

-- Checks only current view condition

-- =====

CREATE VIEW vw_local_check_example AS

SELECT emp_id, emp_name, salary, status

FROM vw_active_employee_check

WHERE salary >= 700000

WITH LOCAL CHECK OPTION;

```
212 • SELECT * FROM vw_local_check_example;
```

emp_id	emp_name	salary	status
1	Aung Aung	850000.00	Active
2	Su Su	1000000.00	Active
4	Hla Hla	1200000.00	Active

```
INSERT INTO vw_local_check_example
```

```
VALUES(6,'Mg Mg Aung', 760000, 'Active');
```

```
217 • SELECT * FROM vw_local_check_example;
```

emp_id	emp_name	salary	status
1	Aung Aung	850000.00	Active
2	Su Su	1000000.00	Active
4	Hla Hla	1200000.00	Active
6	Mg Mg Aung	760000.00	Active

```
219 • SELECT * FROM vw_active_employee_check;
```

emp_id	emp_name	salary	status
1	Aung Aung	850000.00	Active
2	Su Su	1000000.00	Active
4	Hla Hla	1200000.00	Active
5	Mg Mg	600000.00	Active
6	Mg Mg Aung	760000.00	Active

```
-- =====
```

```
-- 12. CREATE OR REPLACE VIEW
```

```
-- Modify existing view without dropping manually
```

```
-- =====
```

```
CREATE OR REPLACE VIEW vw_employee_basic AS
```

```
SELECT emp_id, emp_name, gender, salary
```

```
FROM employees;
```

229 • **SELECT * FROM** vw_employee_basic;

Result Grid				
Filter Rows:				
Export:  Wrap O				
	emp_id	emp_name	gender	salary
▶	1	Aung Aung	Male	850000.00
	2	Su Su	Female	1000000.00
	3	Kyaw Kyaw	Male	700000.00
	4	Hla Hla	Female	1200000.00
	5	Mq Mq	Male	600000.00
	6	Mq Mq Aung	NULL	760000.00

-- =====

-- 13. ALGORITHM = MERGE

-- MySQL tries to merge view query into outer query

-- Good for simple views

-- =====

CREATE OR REPLACE

ALGORITHM = MERGE

VIEW vw_merge_employee AS

SELECT emp_id, emp_name, salary

FROM employees

WHERE salary > 700000;

244 • **SELECT * FROM** vw_merge_employee;

Result Grid			
Filter Rows:			
Export:  Wri			
	emp_id	emp_name	salary
▶	1	Aung Aung	850000.00
	2	Su Su	1000000.00
	4	Hla Hla	1200000.00
	6	Mq Mq Aung	760000.00

```
-- =====
-- 14. ALGORITHM = TEMPTABLE
-- MySQL creates temporary table for view result
-- Usually read-only
-- =====
```

CREATE OR REPLACE

ALGORITHM = TEMPTABLE

VIEW vw_temp_department_summary AS

SELECT dept_id, COUNT(*) AS total_employee

FROM employees

GROUP BY dept_id;

259 • **SELECT * FROM** vw_temp_department_summary;

	dept_id	total_employee
	NULL	1
	1	1
	2	2
	3	1
	4	1

```
-- =====
-- 15. ALGORITHM = UNDEFINED
-- MySQL decides best algorithm
-- =====
```

CREATE OR REPLACE

ALGORITHM = UNDEFINED

VIEW vw_undefined_employee AS

```
SELECT emp_id, emp_name, salary
```

```
FROM employees;
```

```
272 • SELECT * FROM vw_undefined_employee;
```

Result Grid	Filter Rows:	Export:	Wrap Cell C
emp_id	emp_name	salary	
1	Aung Aung	850000.00	
2	Su Su	1000000.00	
3	Kyaw Kyaw	700000.00	
4	Hla Hla	1200000.00	
5	Mq Mq	600000.00	
6	Mq Mq Aung	760000.00	

```
-- =====
```

```
-- 16. SQL SECURITY DEFINER
```

```
-- View runs using permissions of creator
```

```
-- =====
```

```
CREATE OR REPLACE
```

```
SQL SECURITY DEFINER
```

```
VIEW vw_security_definer AS
```

```
SELECT emp_id, emp_name, salary
```

```
FROM employees;
```

```
285 • SELECT * FROM vw_security_definer;
```

Result Grid	Filter Rows:	Export:	Wrap Cell C
emp_id	emp_name	salary	
1	Aung Aung	850000.00	
2	Su Su	1000000.00	
3	Kyaw Kyaw	700000.00	
4	Hla Hla	1200000.00	
5	Mq Mq	600000.00	
6	Mq Mq Aung	760000.00	

-- =====

-- 17. SQL SECURITY INVOKER

-- View runs using permissions of user who calls it

-- =====

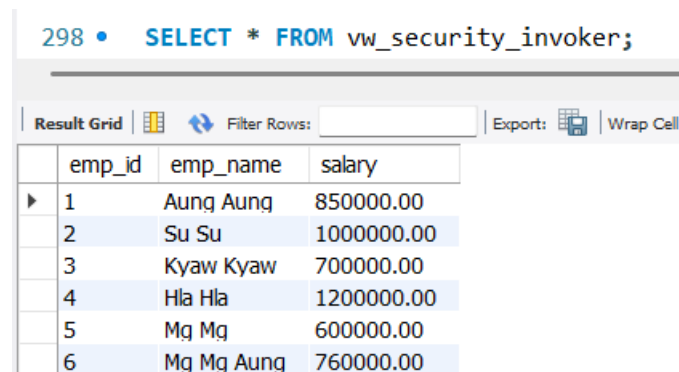
CREATE OR REPLACE

SQL SECURITY INVOKER

VIEW vw_security_invoker AS

SELECT emp_id, emp_name, salary

FROM employees;



298 • SELECT * FROM vw_security_invoker;

Result Grid | Filter Rows: | Export: | Wrap Cell

	emp_id	emp_name	salary
▶	1	Aung Aung	850000.00
	2	Su Su	1000000.00
	3	Kyaw Kyaw	700000.00
	4	Hla Hla	1200000.00
	5	Mq Mq	600000.00
	6	Mq Mq Aung	760000.00

-- =====

-- 18. VIEW FOR DATA SECURITY

-- Hide sensitive columns

-- Example: users can see employee name but not salary

-- =====

CREATE VIEW vw_employee_public AS

SELECT emp_id, emp_name, gender, status

FROM employees;

```
310 • SELECT * FROM vw_employee_public;
```

Result Grid				
Filter Rows:				
Export: Wrap				
	emp_id	emp_name	gender	status
▶	1	Aung Aung	Male	Active
	2	Su Su	Female	Active
	3	Kyaw Kyaw	Male	Inactive
	4	Hla Hla	Female	Active
	5	Mq Mq	Male	Active
	6	Mq Mq Aung	NULL	Active

```
-- =====  
  
-- 19. VIEW FOR REPORTING  
  
-- Monthly or management reporting style  
  
-- =====
```

```
CREATE VIEW vw_management_report AS
```

```
SELECT
```

```
    d.dept_name,
```

```
    COUNT(e.emp_id) AS employee_count,
```

```
    ROUND(AVG(e.salary), 2) AS avg_salary
```

```
FROM departments d
```

```
LEFT JOIN employees e
```

```
ON d.dept_id = e.dept_id
```

```
GROUP BY d.dept_name;
```

```
327 • SELECT * FROM vw_management_report;
```

Result Grid			
Filter Rows:			
Export: Wrap Cell			
	dept_name	employee_count	avg_salary
▶	HR	1	850000.00
	IT	2	850000.00
	Finance	1	1200000.00
	Sales	1	600000.00

-- =====

-- 20. READ-ONLY VIEW EXAMPLES

-- These views cannot normally be updated:

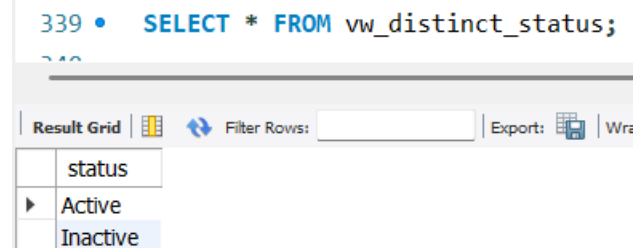
-- GROUP BY, DISTINCT, JOIN, UNION, aggregate functions

-- =====

```
CREATE VIEW vw_distinct_status AS
```

```
SELECT DISTINCT status
```

```
FROM employees;
```



-- This will fail because aggregate view is read-only

```
UPDATE vw_department_salary_summary
```

```
SET average_salary = 1000000
```

```
WHERE dept_name = 'IT';
```

-- =====

-- 21. SHOW VIEW INFORMATION

-- =====


```
350 • SHOW FULL TABLES WHERE Table_type = 'VIEW';
351
```

Result Grid Filter Rows: Export: Wrap Cell Content:		
Tables_in_view_demo	Table_type	
vw_active_employee_check	VIEW	
vw_active_employees	VIEW	
vw_department_salary_summary	VIEW	
vw_distinct_status	VIEW	
vw_employee_basic	VIEW	
vw_employee_department	VIEW	
vw_employee_public	VIEW	
vw_employee_salary_update	VIEW	
vw_high_paid_active	VIEW	
vw_high_salary_employees	VIEW	
vw_local_check_example	VIEW	
vw_management_report	VIEW	
vw_merge_employee	VIEW	
vw_security_definer	VIEW	
vw_security_invoker	VIEW	
vw_temp_department_summary	VIEW	
vw_undefined_employee	VIEW	

```
352 • SHOW CREATE VIEW vw_employee_basic;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
View	Create View	character_set_client	collation_connection	
vw_employee_basic	CREATE ALGORITHM=UNDEFINED D...	utf8mb4	utf8mb4_0900_ai_ci	

```
-- =====
```

```
-- 22. DROP VIEW
```

```
-- =====
```

```
DROP VIEW IF EXISTS vw_employee_public;
```

```
-- =====
```

```
-- END OF SCRIPT
```

```
-- =====
```